Todd Morrill

PhD Student, Computer Science · Columbia University

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EDUCATION

Columbia University — 2024 – present (expected May 2028)
PhD, Computer Science — New York, NY
Advisor: Richard Zemel

Columbia University — Dec 2023
MS, Computer Science — New York, NY
Advisors: Richard Zemel, Kathleen McKeown

Harvard Extension School — May 2021
Computer Science coursework — Online

George Washington University — May 2014
BBA, Finance; minors in Economics and Chinese — Washington, DC

Middlebury College — Summer 2011
Chinese Summer Language Program — Middlebury, VT

RESEARCH EXPERIENCE

Columbia University — Sep 2024 – present
PhD Student, Prof. Richard Zemel, Computer Science — New York, NY

- Derived DeltaNet-PC, a test-time sequence memory model for predictive coding (under review) [P1].
- Lead the NSF AI Institute for Artificial and Natural Intelligence (ARNI) Biological Learning Working Group, focused on benchmarking bio-plausible learning procedures on multimodal / hypermodal data; direct 6 PhD students, mentor masters/undergraduate students, set the group's research agenda, develop the codebase, and report progress in talks and written updates to ARNI leadership and the NSF.
- Developed mixture-of-experts survival models with Columbia University Irving Medical Center clinicians that cluster similar patients while cutting calibration error by up to 70% and improving concordance [C2, W1].

Cold Spring Harbor Laboratory — May 2024 – present
NeuroAI Research Intern & Visiting Scholar, Prof. Anthony Zador, Neuroscience — Cold Spring Harbor, NY

- Published *Bullet Trains*, a first-author ICML 2026 paper, and open-sourced our JAX system that trains event-based SNNs up to 44x faster than sequential baselines while preserving exact neuron dynamics [C1].

Columbia University — Sep 2022 – May 2024
Graduate Research Assistant (MS), Profs. Richard Zemel, Kathleen McKeown — New York, NY

- Co-developed *Prompt Risk Control*, a framework for bounding the deployment risk of large language models, in Prof. Richard Zemel's lab; published at ICLR 2024 [C4, W2].
- Developed a circumplex-theory approach to dialogue analysis for the DARPA Computational Cultural Understanding program in Prof. Kathleen McKeown's lab; oral presentation at LREC-COLING 2024 [C3].

TEACHING EXPERIENCE

Columbia University — Fall 2026
Teaching Assistant, Neural Networks & Deep Learning (Prof. Richard Zemel) — New York, NY

- Guest lectured to a class of 150, designed assignments, and supervised student research projects.

Columbia University — Spring 2026
Teaching Assistant, Continual Learning & Memory Models (Prof. Richard Zemel) — New York, NY

- Lectured on neural memory models to a class of 50, moderated weekly in-class paper discussions, and

supervised 6 student research projects.

RESEARCH INTERESTS

Machine learning and deep learning; memory models (e.g., linear attention); continual learning; biologically plausible learning and credit assignment; spiking neural networks; unsupervised and multimodal representation learning; neuroscience- and cognitive-science-inspired machine learning.

PROFESSIONAL EXPERIENCE

PricewaterhouseCoopers (PwC) Aug 2014 – Mar 2024
Data Scientist, Senior Manager New York, NY

Large Language Model Development

- Built PwC's in-house LLM pipeline in 2023, from FSDP training of Llama 2 (7B–13B) on a 16-GPU A100 cluster through inference benchmarking, load testing, and a hosting cost model for Falcon 40B-class models; first author of the resulting technical report [R1].

Automated Knowledge Graph Construction

- Implemented and compared 7 methods for extracting entities and relations from arbitrary text; accepted to the AKBC 2021 Workshop on Unstructured and Structured Knowledge Bases [W3]. Co-inventor on 5 patents (2 granted) [Pat3–Pat7].

Information Retrieval and Classification for a Bank

- Managed a team of 6 and collaborated with CMU (Eduard Hovy) to build models that classify regulatory text for compliance purposes.
- Lead inventor on 2 patents (1 granted) [Pat1, Pat2]; reduced time spent by compliance experts by up to 50% [bit.ly/32f2lx3].

Enterprise Search

- Reduced the number of clicks required for 50K+ users to find information in PwC's firm-wide search portal; presented at Google Cloud Next '18 [youtu.be/63EANkPzuJY].

Selected Projects Led

- Domain-adapting large language models; an EEG-based brain–computer interface [youtu.be/f81T0KcprpM]; deep learning on edge devices [youtu.be/iqVKjye-J68]; structural causal modeling with CMU students; dynamic pricing for an online ticketing platform; and forecasting, classification, and information-extraction models for technology, insurance, automotive, food service, and financial services clients.

PEER-REVIEWED CONFERENCE PAPERS

[C1] **Todd Morrill**, Christian Pehle, Anthony Zador. Bullet Trains: Parallelizing Training of Temporally Precise Spiking Neural Networks. *International Conference on Machine Learning (ICML)*, 2026.

[C2] **Todd Morrill**, Aahlad Manas Puli, Murad Megjhani, Soojin Park, Richard Zemel. Let the Experts Speak: Improving Survival Prediction & Calibration via Mixture-of-Experts Heads. *Machine Learning for Health (ML4H) Symposium*, 2025.

[C3] **Todd Morrill**, Zhaoyuan Deng, Yanda Chen, Amith Ananthram, Colin Wayne Leach, Kathleen McKeown. Social Orientation: A New Feature for Dialogue Analysis. *LREC-COLING*, 2024. Oral presentation.

[C4] Thomas Zollo, **Todd Morrill**, Zhun Deng, Jake Snell, Toniann Pitassi, Richard Zemel. Prompt Risk Control: A Rigorous Framework for Responsible Deployment of Large Language Models. *International Conference on Learning Representations (ICLR)*, 2024.

PEER-REVIEWED WORKSHOP PAPERS

[W1] **Todd Morrill**, Aahlad Manas Puli, Murad Megjhani, Soojin Park, Richard Zemel. Let the Experts Speak: Improving Survival Prediction & Calibration via Mixture-of-Experts Heads. *Learning from Time Series for Health Workshop, NeurIPS*, 2025.

[W2] Thomas Zollo, **Todd Morrill**, Zhun Deng, Jake Snell, Toniann Pitassi, Richard Zemel. Prompt Risk Control: A Rigorous Framework for Responsible Deployment of Large Language Models. *Socially Responsible Language Modelling Research (SoLaR) Workshop, NeurIPS*, 2023.

[W3] **Todd Morrill**. Cluster Based Named Entity Recognition. *Workshop on Unstructured and Structured Knowledge Bases, AKBC*, 2021.

PREPRINTS

[P1] **Todd Morrill**, Richard Zemel. DeltaNet-PC: Test-Time Memory Without Backpropagation Through Time. arXiv:2610.15185, 2026. [arxiv.org/abs/2610.15185]

TECHNICAL REPORTS

[R1] **Todd Morrill**, Michael Kirchner, Vasudeva Sankaranarayanan, Joseph Voyles. Technical Report: Large Language Models. 2023.

[R2] **Todd Morrill**, Satyam Sharma. Comparing Hyperdimensional Computing to Deep Learning for Natural Language Processing Tasks. 2023.

PATENTS

[Pat1] **Todd Morrill**, Eric Roma, Nicolas Kuzak, Neelam Sharma, Andrew Runge, Jayvardhan Rathi, Waqar Sarguroh, Wenting Zhao. Regulatory Tree Parser. US Patent 12,072,861, granted 2024 (filed 2022).

[Pat2] **Todd Morrill**, Eric Roma, Neelam Sharma, Alistair Moore, Andrew Runge. Regulatory Obligation Identifier. US Patent App. 2022/0374914, pending (filed 2022).

[Pat3] Paul Sheward, Chung-Sheng Li, Scott Likens, and 9 others incl. **Todd Morrill**. Method and Apparatus for an AI-Assisted Virtual Consultant. US Patent 12,282,501, granted 2025.

[Pat4] Paul Sheward, Chung-Sheng Li, Scott Likens, and 9 others incl. **Todd Morrill**. Method and Apparatus for Knowledge Representation and Reasoning in Accounting. US Patent 12,229,195, granted 2025.

[Pat5] Paul Sheward, Chung-Sheng Li, Scott Likens, and 9 others incl. **Todd Morrill**. Systems and Methods for Constructing Topic-Specific Knowledge Graphs. US Patent App. 2024/0394563, pending.

[Pat6] Paul Sheward, Chung-Sheng Li, Scott Likens, and 9 others incl. **Todd Morrill**. Method and Apparatus for an AI-Assisted Virtual Assistant. US Patent App. 2024/0403302, pending.

[Pat7] Paul Sheward, Chung-Sheng Li, Scott Likens, and 9 others incl. **Todd Morrill**. Method and Apparatus for AI-Assisted Virtual Assistant for SME Agent. US Patent App. 2025/0028742, pending.

TALKS AND PRESENTATIONS

1. Memory Models in Artificial and Biological Systems. Entrypoint Capital, 2026.
2. Local Circuits: Memory, Neuroscience and AI. Zuckerman Institute, Columbia University, 2026.
3. Machine Learning Enabled Chatbots. Google Cloud Next, 2018.
4. Machine Learning Enabled Chatbots. AI4 Finance, 2018.

TECHNICAL SKILLS

Programming languages	Python, C/C++ (+CUDA), JavaScript, SQL, Bash, LaTeX, Markdown
ML libraries	JAX, PyTorch, TensorFlow/Keras, scikit-learn, pandas, NumPy, SciPy
Tools	SLURM, Docker, Kubernetes, Git/GitHub, VS Code
Cloud	Google Cloud, Amazon Web Services, Microsoft Azure
Spoken languages	Professional proficiency in Mandarin Chinese

LEADERSHIP AND SERVICE

- Midnight Run NYC volunteer, 2024 – present
- Harlem Grown volunteer, 2014 – 2020
- GWU Alternative Breaks volunteer, rebuilding after Hurricane Katrina, 2011 – 2013
- Eagle Scout, Boy Scouts of America, 2010